

Electrical AC BOM (Phase 1)

As-of date: 2026-09-02

Purpose: maintain the installed Phase 1 AC architecture baseline: portable 30A EMS, 30A shore inlet/locking cord with modular 25 ft TT-30 extension, one 6-way DIN enclosure, 30A AC-in breaker, 30A AC-out main breaker, and two active 20A GFCI-protected AC-out branches. Owner report 2026-09-02 : all 120V work is in enclosures, including four duplex devices/eight plug positions total, and the system has seen regular use. Physical enclosure count is closed. Final safety acceptance still requires explicit LINE/LOAD/PE proof, downstream protection, GFCI trip/reset in inverter and accepted-shore modes, polarity/neutral-isolation checks, the MultiPlus external PE/chassis and separate shell bonds, and representative-load verification.

Related docs:

- docs/implementation/ELECTRICAL_overview_diagram.md
- docs/core/SYSTEMS.md
- docs/core/TRACKING.md
- bom/bom_estimated_items.csv

Historical provenance

- [INSTALL MINUS 12 READINESS PLAN](#) preserves the May 7 install-window AC planning context; this file owns current Phase 1 AC procurement/implementation.

Historical physical shore-inlet install gate (2026-07-19) — closed/superseded by as-built state

- The L5-30 inlet was then purchased but not yet cut into the camper/bed-side interface.
- The electrical module was hard-mounted. The MultiPlus and combined AC breaker enclosure had been removed for the lift and still required remounting at that time.
- From the remounted AC-in breaker endpoint, mock the full 10/3 path to the candidate exterior inlet with required bend radius, cable support, drip/water management, service loop, strain relief, and access to both sides of the cut.
- Any small cable opening through the plywood electrical backer requires a correctly sized grommet/bushing or gland plus independent cable support/strain relief; bare 10/3 must not bear on a raw plywood edge.
- Confirm the exterior cover swing, connector clearance, wall/backing thickness, hidden structure/no-drill zone, butyl bedding land, finish-seal geometry, and a path that does not share the wet-side chase.
- Cut only after the complete inside and outside route is proven. Do not let the easiest exterior location create an inaccessible cable entry or service-obscuring bend behind the electrical module.

Locked AC Architecture

AC-in chain (shore to inverter)

- shore source/adapters -> portable 30A EMS -> optional 25 ft TT-30 extension -> 30A locking shore cord -> L5-30 shore inlet -> combined 6-way AC DIN enclosure -> 30A UL489 AC-in breaker/disconnect -> MultiPlus AC-in (L/N/PE)
- AC-in conductors are 10 AWG / 10/3 on the protected AC-in path (30A hardware basis).
- MultiPlus input current limit is set to actual source when adapters are used. Use 10A for first household tests and 12A maximum policy on a normal 15A outlet; do not leave the limit at 50A on adapter/household shore.
- Use the modular extension only when the extra reach is needed. Fully seat, elevate, and weather-protect the TT-30 midpoint connection, and uncoil both cords under load.

AC-out chain (inverter-backed branch distribution)

- MultiPlus AC-out-1 -> 10/3 feeder -> combined 6-way AC DIN enclosure -> 30A UL489 AC-out main breaker -> 20A Branch A + 20A Branch B -> GFCI receptacle per branch
- The two first-in-chain receptacles are 20A self-test GFCIs: Branch A = office/driver side, Branch B = Galley/passenger side. Owner report 2026-09-02 : each GFCI feeds one downstream duplex from its LOAD side, all four devices are enclosed, and AC has seen regular use. Formal electrical acceptance remains open only for the explicit safety/functional checks below.
- Do not reopen the box-count task. Confirm 12 AWG , LINE/LOAD/PE , PE continuity, labels, GFCI trip/reset, whole-chain downstream protection, polarity/neutral isolation, and representative loads in inverter and accepted-shore modes.
- AC-out branch hardware is not required to perform the initial AC-in-only battery charging test, but it is now included in the purchased Phase 1 cart.

Neutral and ground handling

- AC-in and AC-out neutral termination paths remain isolated.
- Separate AC-in neutral pass-through/termination and AC-out neutral bar are required inside the combined enclosure.
- Common equipment grounding/PE bus is acceptable; do not use it as a neutral.
- Continuous equipment ground path and chassis bond are required end-to-end. The installed MultiPlus-II 48/3000/35-50 120V is a Class I device; its official manual identifies the external M6 connection as the primary PE point, requires at least 4 mm² grounding conductor, and explicitly requires the casing to be connected to the vehicle chassis in a mobile shore-power installation. Use 10 AWG green stranded copper from the MultiPlus external M6 PE lug to a verified truck-chassis bond point; do not leave the lug open and do not use the aluminum Hiatus shell or 80/20 as the only protective-earth path.
- Bond the aluminum camper shell separately into the same chassis/equipment-ground network with a corrosion-compatible connection. Treat shell hardware/body contact as unproven until low-resistance continuity is measured; use compatible/tinned terminals and antioxidant/isolation practice appropriate to the aluminum interface.
- Do not jumper the MultiPlus case/PE lug to Lynx negative or the 12V negative bus. The 48V and 12V DC return paths remain on their dedicated conductors/buses; Mechman case-ground behavior is a separate commissioning check that may establish the single deliberate house-negative-to-chassis reference through the alternator and its dedicated 2/0 return.
- Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring. Leave the MultiPlus internal ground relay enabled for the normal single-unit topology: it bonds AC-out neutral to chassis in inverter mode and opens that bond when shore AC is accepted.

AC-out-2 policy

- AC-out -2 is **reserve-only** in Phase 1.
- Keep labeled panel space and capped route only; no energized branch hardware is procured for this path in Phase 1.

Required Purchasable Components (Phase 1)

Immediate AC-in-only initial charge path

These rows unblock safe MultiPlus shore charging and should not wait on final receptacle count:

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord, modular extension + household adapter	1 locking cord + 1 extension + 1 dogbone	30A TT-30P-to-L5-30R cord, 25 ft TT-30P-to-TT-30R extension, and 15A household dogbone adapter	108, 340, 281	Purchased
Portable EMS/surge protector	1	Portable 30A EMS with open-neutral/polarity/voltage protection	123	Purchased
AC-in breaker/disconnect	1 of 2	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
Combined AC DIN enclosure and bus hardware	1 enclosure + accessories	6-way DIN enclosure; physically isolate AC-in and AC-out neutrals	109, 14	Purchased
Shore + AC-in feed cable	as routed	10/3 stranded tinned-copper cable for C-28/C-29 (30A path)	114	Purchased
Shore-inlet seal/bedding consumables	per install	Butyl bedding + compatible exterior polyurethane perimeter/finish seal	179, 180	Purchased
Strain relief/grommets/clamps/labels/ferrules	per entries	Sized for selected cable/enclosure terminals	38, 41, 43, 44, 45	Cable glands on hand; ferrule kit purchased

Full AC-out distribution / final Phase 1 branch hardware

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased

Shore cord, modular extension + household adapter	1 locking cord + 1 extension + 1 dogbone	30A TT-30P-to-L5-30R cord, 25 ft TT-30P-to-TT-30R extension, and 15A household dogbone adapter	108, 340, 281	Purchased
Portable EMS/surge protector	1	Portable 30A EMS used source-side before the shore cord	123	Purchased
AC-in breaker/disconnect	1	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
AC-out main breaker	1	DIN-mount UL 489 1-pole 30A 120VAC, ahead of branch breakers	327	Purchased
Combined DIN enclosure	1	6-way DIN enclosure with slot plan: 30A in, 30A out, 20A, 20A, 2x blank/spare	109	Purchased
AC-out branch breaker set	2 active	DIN-mount UL 489 branch breakers: 20A + 20A	110, 330	Purchased
DIN accessory kit	1 kit	Neutral isolation hardware, PE/ground bus support, ferrules, labels/blanks as needed	14, 41	Purchased / labels-blanks confirm on hand
MultiPlus/chassis and shell bonding materials	field-measured	10 AWG green stranded copper minimum for MultiPlus M6 PE to verified truck chassis; separate shell bond jumper; M6/tinned lugs, protected chassis attachment, compatible aluminum-interface hardware/compound	51, 339	Inventory/buy
GFCI receptacles	2	20A self-test GFCI receptacles, one per active branch	15	Purchased
Standard downstream receptacles	2 installed / owner-reported	One downstream duplex from each first GFCI LOAD, same 12 AWG branch, accessible listed box	inactive historical 111; installed inventory source not yet reconciled	Inspect/test as-built
Outlet boxes + covers/faceplates + clamps	4 installed device sets / owner-reported	Six 14 cu in old-work boxes were purchased; prove each used device's listed box, compatible cover/plate and clamp, conductor fill, wall-stack fit, and service access	active row 112 plus inventory	Physical acceptance open
AC branch cable	30 ft purchased	12/3 stranded triplex branch cable (C-31/C-32)	113	Purchased
Shore + AC-in/AC-out feeder cable	20 ft purchased	10/3 stranded triplex for C-28/C-29/C-30 (30A paths)	114	Purchased
Strain relief/cable glands	per enclosure entries	Use assorted on-hand entry hardware sized during physical layout	44	On-hand fitment stock
Grommets	per pass-through points	Abrasion protection at penetrations, selected hands-on during routing	43	On-hand / install fit
P-clamps and retention hardware	per route	Cable support and vibration control, selected hands-on during routing	45	On-hand / install fit
Loom/sleeving	per exposed runs	Harness abrasion protection	42	Required
Heat shrink (adhesive)	install consumable	Termination sealing and strain relief support	38	Required
Ferrules/terminals (AC-relevant)	install consumable	Sized to 10 AWG and 12 AWG terminations as required by device terminals	41, 116	Required

AC-out-2 branch breaker/protection hardware	ø in Phase 1	Reserve-only route, no energized branch hardware in this phase	N/A	Reserve-only (not procured)
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First-Live AC-In Test Result (2026-05-27)

- AC Input 1 should be labeled Shore power for this mobile/source-current-limited system.
- MultiPlus switch II is charger-only; switch I is inverter/charger normal mode; 0 is off.
- Short shore test passed with household-source current limiting: about 1294W shore input and about 54.3V x 21.6A (~1173W) battery charge in bulk.
- Current disconnect practice: connect RV/EMS/load side first, then energize shore; disconnect in reverse by de-energizing/unplugging shore source before disconnecting the RV side.
- This result does not close battery-charge-profile commissioning. Program/verify the MultiPlus lithium settings before sustained charging.

Manual AC Validation Checklist

0) AC-in-only initial charger validation

- Confirm AC-in physical order: shore source/adapters -> portable EMS -> optional TT-30 extension -> locking shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in .
- Confirm AC-out breakers/loads are disconnected or not yet installed for the first battery-charge test.
- Confirm MultiPlus input current limit is set to actual source (10A first household test, 12A max on normal 15A , actual rating for 20A / 30A).
- Confirm AC Input 1 is labeled/configured as Shore power .
- Confirm battery charge profile is intentionally set for the selected LiFePO4 voltage basis before sustained charging.
- Confirm AC-in and AC-out neutral paths are not mixed.

Use this checklist as the continuing acceptance gate before sustained shore charging, AC-out branch energization, or any AC enclosure closeout.

1) Topology integrity

- Confirm one unique AC-in chain exists: shore source/adapters -> portable EMS -> cord -> inlet -> AC-in breaker -> MultiPlus AC-in .
- Confirm one unique AC-out-1 chain exists: MultiPlus AC-out-1 -> 30A AC-out main -> 20A branch breakers -> GFCI receptacles .
- Confirm AC-out-2 is documented as reserve-only and not active in Phase 1 procurement.
- Confirm the two first-in-chain GFCIs remain the only branch-origin devices and each owner-reported downstream duplex is fed only from its corresponding GFCI LOAD ; prove compatible listed boxes/covers/clamps, conductor fill/accessibility, and whole-chain testing.

2) Protection coordination

- Confirm AC-in breaker is 30A and AC-in conductors are 10 AWG .
- Confirm AC-out main breaker is 30A and the MultiPlus-to-enclosure feeder is 10 AWG .
- Confirm branch OCP values are 20A and 20A with 12 AWG branch conductors.
- Confirm breaker listing basis is UL 489 (or equivalent NRTL listing) for branch/feeder use.

3) Neutral/ground correctness

- Confirm AC-in and AC-out neutral paths are isolated.
- Confirm input and output neutral paths are separately landed/passed through inside the combined enclosure.
- Confirm the equipment grounding/PE path is common and continuous.
- Confirm the MultiPlus external M6 PE lug is bonded to a verified truck-chassis point with at least 4 mm² / selected 10 AWG green stranded copper; the aluminum shell/80/20 is not the sole return.
- Confirm the aluminum shell has its own corrosion-compatible bond to the chassis/equipment-ground network and verify low-resistance continuity after final assembly.
- Confirm no intentional jumper exists from MultiPlus case/PE to Lynx negative or the 12V negative bus, and confirm no chassis path bypasses the SmartShunt after the Mechman ground style is physically identified.
- Confirm no fixed downstream neutral-ground bond is added in branch receptacle wiring.
- Confirm the MultiPlus internal ground relay remains enabled for this normal single-unit mobile topology; test both GFCI branches in inverter mode and on accepted shore power before routine use.

4) Procurement completeness

- Confirm every required AC component class has a BOM row mapping.
- Confirm no AC-critical component exists only as implied text.

- Confirm AC-out-2 hardware remains excluded from Phase 1 carts.

5) Documentation parity

- Confirm AC assumptions match across:
 - docs/implementation/ELECTRICAL_AC_BOM.md
 - docs/implementation/ELECTRICAL_overview_diagram.md
 - docs/core/SYSTEMS.md
 - docs/core/TRACKING.md
 - bom/bom_estimated_items.csv

6) Operating scenarios

- AC-in-only initial charge: no AC-out loads connected, MultiPlus charger behavior documented.
- Shore present (30A source): pass-through + charging behavior documented.
- Shore present via 15A/20A adapter: current-limit setting policy documented.
- Shore absent: inverter-backed AC-out-1 behavior documented.
- GFCI trip/reset behavior per branch is called out for commissioning test.

Procurement Notes

- DIN rail is a mounting method; breaker listing and rating remain the controlling requirement.
- Lowest-cost listed policy is acceptable only if each selected device has verifiable NRTL listing (UL or ETL) for intended use.
- Purchased SKU lock is recorded in bom/bom_estimated_items.csv for rows 13 , 327 , 14 , 15 , 41 , 107 , 108 , 109 , 110 , 112 , 330 , 113 , 114 , 123 , 179 , and 180 . Rows 13 / 327 are the AC-in/AC-out 30A pair; rows 110 / 330 are the two 20A branch breakers. Row 112 is the purchased old-work box six-pack, not evidence that six downstream devices are approved. Row 181 was same-order tire-deflator/off-road support hardware and is intentionally outside AC scope.
- Inactive BOM row 111 preserves the formerly closed downstream-receptacle concept. Row 112 supplied the six purchased old-work boxes. Owner report 2026-08-27 closes physical device count at two GFCIs plus two downstream duplexes; installed receptacle/cover/clamp inventory provenance and all box-fill/wall-stack/accessibility/testing gates remain open.
- Current utilization note (2026-05-16): AC protection chain and purchase scope are locked for a 30A system with two active 20A branches. Do not add a third active AC branch without revisiting the AC-out main/enclosure/feed plan.